



# Variable Acceleration · Mark Scheme

Mechanics Chapter 4: Variable Acceleration. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

## Section A – Differentiating

- 1  $v = 8t - 3$  M1;  $v = 13 \text{ ms}^{-1}$  A1.
- 2  $a = 3t^2 - 2$  M1;  $a = 1 \text{ ms}^{-2}$  A1.
- 3  $v = 6t^2 - 18t + 12$  M1;  $= 6(t-1)(t-2) = 0$  M1;  $t = 1$  and  $t = 2$  A1.
- 4 (a)  $a = 6t - 12 = 6 \text{ ms}^{-2}$  B1. (b)  $a = 0$  at  $t = 2$  M1;  $v = 12 - 24 = -12$ , speed  $12 \text{ ms}^{-1}$  A1.
- 5(a)  $v = 3t^2 - 12$  B1.
- 5(b)  $3t^2 = 12$  M1;  $t = 2$  A1;  $x = 8 - 24 + 3 = -13$  m A1.
- 5(c)  $a = 6t = 12 \text{ ms}^{-2}$  B1.
- 5(d)  $v = 27 - 12 = 15 \text{ ms}^{-1}$  B1.
- 5(e) Moving in the negative direction before  $t = 2$ , and in the positive direction after B1.

## Section B – Integrating

- 6  $v = 3t^2 + c$  M1;  $v = 3t^2 + 2$  A1.
- 7  $s = 2t^2 - \frac{1}{3}t^3$  M1 A1;  $s = 18 - 9 = 9$  m A1.
- 8  $v = 12t - 3t^2 + c$ ,  $c = 0$  M1 A1;  $v = 9 \text{ ms}^{-1}$  A1.
- 9  $v = t^2 - 4t + 3$  M1 A1;  $s = \frac{1}{3}t^3 - 2t^2 + 3t$  A1;  $s(3) = 9 - 18 + 9 = 0$  A1.
- 10(a)  $v = 3t^2 - 4t + c$  M1;  $v = 3t^2 - 4t - 2$  A1.
- 10(b)  $s = t^3 - 2t^2 - 2t + c$  M1;  $c = 0$ , so  $s = t^3 - 2t^2 - 2t$  A1.
- 10(c)  $s(3) = 27 - 18 - 6 = 3$  m B1.
- 10(d)  $s(1) = -3$  M1; displacement  $= 3 - (-3) = 6$  m A1.
- 10(e)  $6t - 4 = 0 \Rightarrow t = \frac{2}{3}$  s B1.

## Section C – Stationary values and distance travelled

- 11  $6 - 2t = 0$  M1;  $t = 3$  s A1.
- 12  $\frac{dv}{dt} = 2t - 6 = 0 \Rightarrow t = 3$  M1;  $v = 9 - 18 + 8 = -1 \text{ ms}^{-1}$  A1; speed  $1 \text{ ms}^{-1}$  A1.
- 13  $v = 3(t-1)(t-3) \geq 0$  for  $0 \leq t \leq 1$ , so no change of direction B1;  $\int_0^1 (3t^2 - 12t + 9) dt = [t^3 - 6t^2 + 9t]_0^1$  M1;  $= 4$  m A1.
- 14 Direction changes at  $t = 1$  B1;  $[t^3 - 6t^2 + 9t]_1^3 = 0 - 4 = -4$  M1; total distance  $4 + 4 = 8$  m A1; displacement 0 B1.
- 15(a)  $\frac{dv}{dt} = 12 - 6t = 0 \Rightarrow t = 2$  M1;  $v = 12 \text{ ms}^{-1}$  A1;  $\frac{d^2v}{dt^2} = -6 < 0$ , so a maximum A1.
- 15(b)  $3t(4-t) = 0 \Rightarrow t = 4$  B1.
- 15(c)  $[6t^2 - t^3]_0^4$  M1;  $= 96 - 64 = 32$  m A1.
- 15(d)  $v < 0$  for  $t > 4$ , so split at  $t = 4$  M1;  $[6t^2 - t^3]_4^6 = 0 - 32 = -32$  A1; total distance 64 m A1.



## Roar-level challenge

**16**  $v = (t-1)(t-4)$ , direction changes at  $t = 1$  and  $t = 4$  *M1*;  $s = \frac{1}{3}t^3 - \frac{5}{2}t^2 + 4t$  *M1*;  $s(1) = \frac{11}{6}$ ,  $s(4) = -\frac{8}{3}$ ,  $s(5) = -\frac{5}{6}$  *A1*; distance  $\frac{11}{6} + \frac{27}{6} + \frac{11}{6}$  *M1*;  $= \frac{49}{6} = 8.17$  m *A1*.